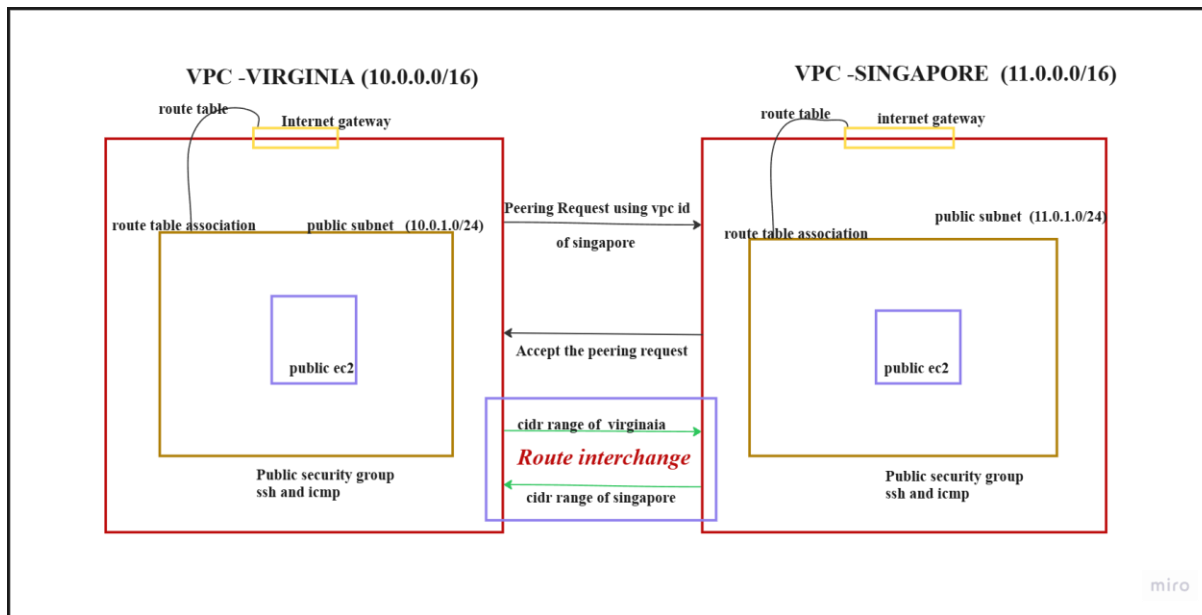


VPC PEERING

VPC Peering is a networking connection between two Amazon Virtual Private Cloud (VPCs) that allows them to communicate with each other using private IP addresses. It is a service provided by Amazon Web Services that enables resources like Amazon EC2 instances in different VPCs to communicate securely as if they were in the same network.



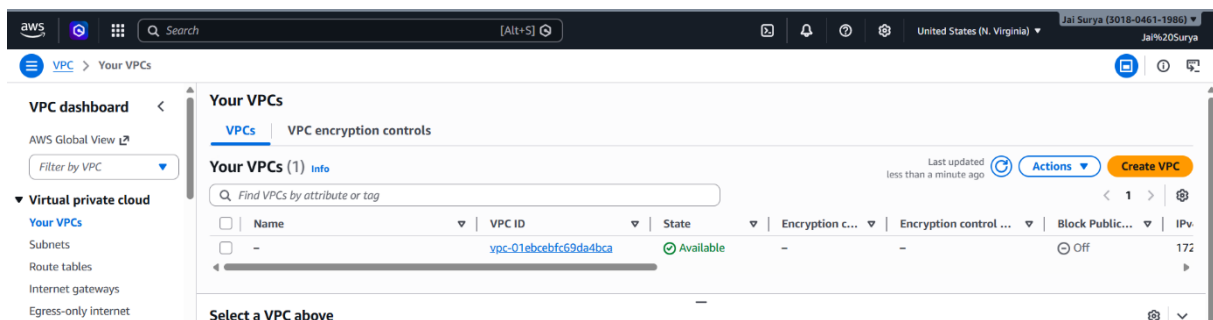
HANDS-ON FOR VPC PEERING

Screenshot 1 – Create VPC in N. Virginia.

Step 1: Change region to N. Virginia.

Step 2: Go to VPC service.

Step 3: Click Create VPC.



Screenshot 2 – VPC Details

Step 1: Enter VPC name.

Step 2: Set IPv4 CIDR to 10.0.0.0/16.

Step 3: Click Create VPC.

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

vpc-virg

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

Screenshot 3 – VPC Created

Step 1: Click Create VPC.

Step 2: VPC is created successfully.

VPC dashboard [Your VPCs](#)

Virtual private cloud

Your VPCs

Subnets

Route tables

Subnets (2) [Info](#)

Find VPCs by attribute or tag

Last updated less than a minute ago

Actions [Create VPC](#)

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR
☐	-	vpc-01ebcebf69da4bca	Available	-	-	Off	172.31.0.0/20
☐	vpc-virg	vpc-04cc0c09433b34e42	Available	-	-	Off	10.0.0.0/16

Screenshot 4 – Creating Subnet in Virginia

Step 1: Go to Subnets.

Step 2: Click Create subnet.

VPC dashboard [Your VPCs](#)

Virtual private cloud

Subnets

Route tables

Internet gateways

Egress-only internet

Subnets (6) [Info](#)

Find subnets by attribute or tag

Last updated 5 minutes ago

Actions [Create subnet](#)

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	-	subnet-067ad608b51a3183d	Available	vpc-01ebcebf69da4bca	Off	172.31.80.0/20
☐	-	subnet-0bc21adafb4a57e2	Available	vpc-01ebcebf69da4bca	Off	172.31.16.0/20
☐	-	subnet-0139dea20f453ec9	Available	vpc-01ebcebf69da4bca	Off	172.31.64.0/20
☐	-	subnet-0b7c653d9ae8f0530	Available	vpc-01ebcebf69da4bca	Off	172.31.48.0/20
☐	-	subnet-06995f80244c501e1	Available	vpc-01ebcebf69da4bca	Off	172.31.0.0/20

Select a subnet

Screenshot 5 – Subnet Details

Step 1: Select the VPC.

Step 2: Enter Subnet name.

Step 3: Select the Availability Zone (AZ).

Step 4: Enter Subnet CIDR.

The screenshot shows the 'Create subnet' page in the AWS Management Console. The breadcrumb navigation at the top reads 'VPC > Subnets > Create subnet'. The page title is 'Create subnet' with an 'Info' link. Under the 'VPC' section, the 'VPC ID' is set to 'vpc-04cc0c09433b34e42 (vpc-virg)'. The 'Associated VPC CIDRs' section shows 'IPv4 CIDRs' as '10.0.0.0/16'. The 'Subnet settings' section includes a 'Subnet name' field with 'subnet-virg', an 'Availability Zone' dropdown set to 'United States (N. Virginia) / use1-az1 (us-east-1a)', an 'IPv4 VPC CIDR block' dropdown set to '10.0.0.0/16', and an 'IPv4 subnet CIDR block' text input set to '10.0.1.0/24' with a '256 IPs' indicator.

Screenshot 6 – Subnet Created

Step 1: Click Create subnet.

Step 2: Subnet is created successfully.

The screenshot shows the 'Subnets' page in the AWS Management Console. A green success message at the top states: 'You have successfully created 1 subnet: subnet-0eee099df5a491aae'. The page title is 'Subnets (7)' with an 'Info' link. A search bar is present with the placeholder 'Find subnets by attribute or tag'. Below the search bar is a table of subnets.

	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	-	subnet-06cdef51627a311b3	Available	vpc-01ebcebf69da4bca	Off	172.31.32.0/2
☐	subnet-virg	subnet-0eee099df5a491aae	Available	vpc-04cc0c09433b34e42 vpc...	Off	10.0.1.0/24

Screenshot 7 – Creating Route Table in Virginia

Step 1: Go to Route Tables.

Step 2: Click Create route table.

Step 3: Enter route table details and select the VPC.

Step 4: Click Create route table.

The screenshot shows the AWS Management Console interface for creating a new route table. The top navigation bar includes the AWS logo, a search bar, and the user's profile. The left sidebar shows the 'VPC dashboard' with links to 'Virtual private cloud', 'Your VPCs', 'Subnets', and 'Route tables'. The main content area is titled 'Route tables (1)' and shows a table with one entry: 'rtb-0c5fba2dbeb4ff11c' associated with VPC 'vpc-01ebcebfc69da4bca'. Below this, the 'Create route table' wizard is displayed. It includes a 'Route table settings' section with a 'Name' field (optional) containing 'rt-virg' and a 'VPC' dropdown menu selected to 'vpc-04cc0c09433b54e42 (vpc-virg)'. The 'Tags' section shows a key-value pair 'Name: rt-virg'. At the bottom right, there are 'Cancel' and 'Create route table' buttons.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-0c5fba2dbeb4ff11c	-	-	Yes	vpc-01ebcebfc69da4bca

Create route table

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
rt-virg

VPC
The VPC to use for this route table.
vpc-04cc0c09433b54e42 (vpc-virg)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key
Name

Value - optional
rt-virg

Buttons: Add new tag, Cancel, Create route table

Screenshot 8 – Creating Internet Gateway in Virginia

Step 1: Go to Internet Gateways.

Step 2: Click Create internet gateway.

Step 3: Enter the internet gateway name.

Step 4: Click Create internet gateway.

The screenshot shows the AWS Management Console interface for Internet Gateways. The top navigation bar includes the AWS logo, a search bar, and the user's profile. The left sidebar shows the 'VPC dashboard' with links to 'Virtual private cloud', 'Your VPCs', 'Subnets', and 'Internet gateways'. The main content area is titled 'Internet gateways (1)' and shows a table with one entry: 'igw-01f6b03ff1cf88287' in an 'Attached' state, associated with VPC 'vpc-01ebcebfc69da4bca' and owned by '301804611986'. At the bottom right, there are 'Actions' and 'Create internet gateway' buttons.

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-01f6b03ff1cf88287	Attached	vpc-01ebcebfc69da4bca	301804611986

Buttons: Actions, Create internet gateway

Create internet gateway Info

An internet gateway is a virtual router that connects a VPC to the internet. To create a new internet gateway specify the name for the gateway below.

Internet gateway settings

Name tag
Creates a tag with a key of 'Name' and a value that you specify.

internet-virg

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q internet-virg X Remove

Add new tag

You can add 49 more tags.

Cancel Create internet gateway

Screenshot 9 – Attach Internet Gateway to VPC

Step 1: Select the created Internet Gateway.

Step 2: Click Actions.

Step 3: Click Attach to VPC.

Step 4: Select the Virginia VPC and attach it.

Internet gateways (1/2) Info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID
-	igw-01f6b03ff1cf88287	Attached	vpc-01ebcebf6c9da4bca
internet-virg	igw-0be39f56a2c4ebd92	Detached	-

Actions

- View details
- Attach to VPC
- Detach from VPC
- Manage tags
- Delete internet gateway

Screenshot 10 – Subnet Association

Step 1: Go to Route Tables.

Step 2: Select the created Virginia route table.

Step 3: Click Subnet associations.

Step 4: Click Edit subnet associations.

Route tables (1/3) Info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-0c5fba2dbef4ff11c	-	-	Yes	vpc-01ebcebf6c9da4bca
-	rtb-065d5ce64c4d06fdc	-	-	Yes	vpc-04cc0c09433b34e42 vpc...
rt-virg	rtb-07890aaa2fe81cef0	-	-	No	vpc-04cc0c09433b34e42 vpc...

rtb-07890aaa2fe81cef0 / rt-virg

Details Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (0)

Find subnet association

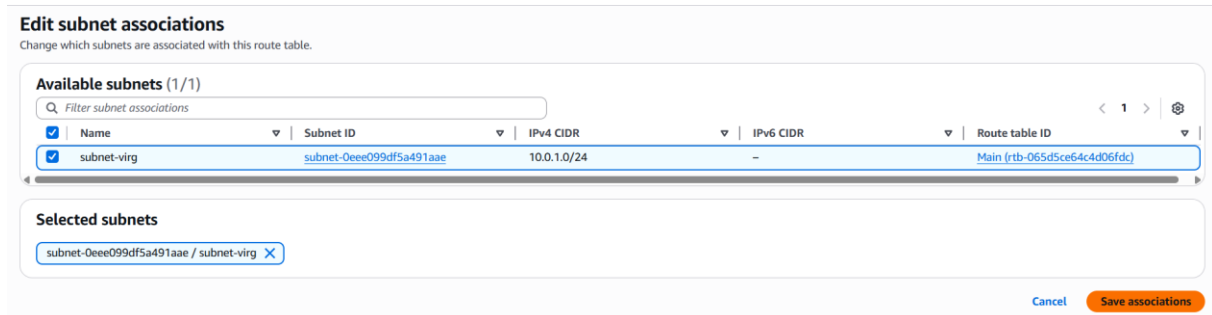
Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
No subnet associations			

You do not have any subnet associations.

Screenshot 11 – Select Subnet

Step 1: Select the subnet.

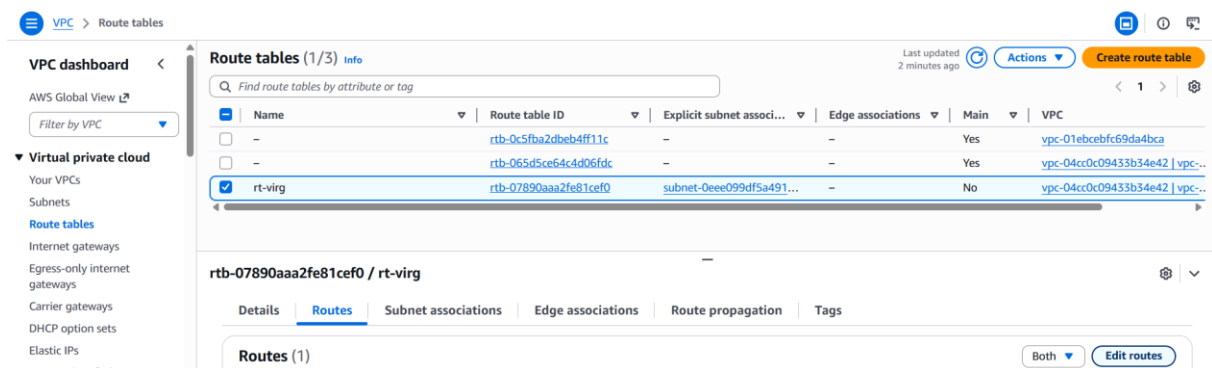
Step 2: Click Save.



Screenshot 12 – Edit Routes

Step 1: Go to Routes in the route table.

Step 2: Click Edit routes.



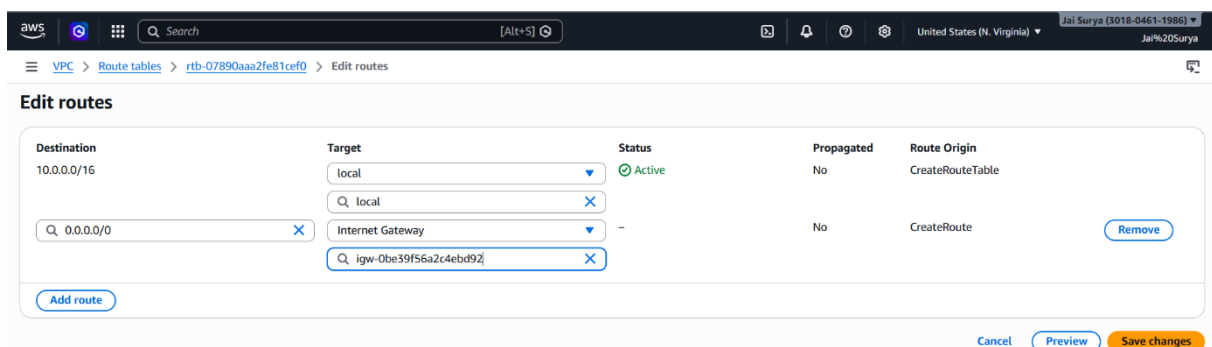
Screenshot 13 – Add Route

Step 1: Set Destination to 0.0.0.0/0.

Step 2: Select Internet Gateway as the target.

Step 3: Choose the created Internet Gateway.

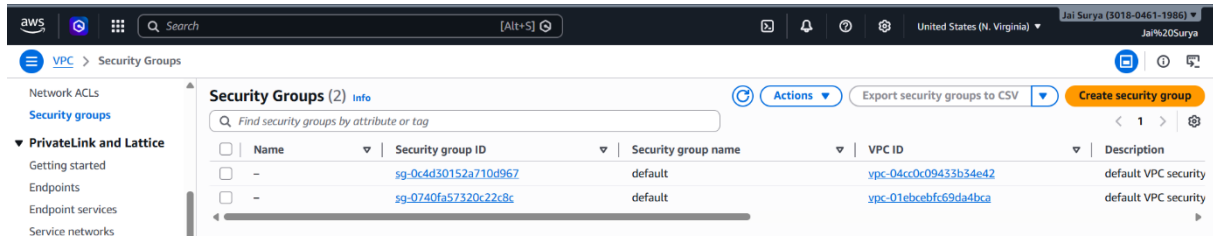
Step 4: Click Save.



Screenshot – Create Security Group

Step 1: Go to Security Groups.

Step 2: Click Create security group.

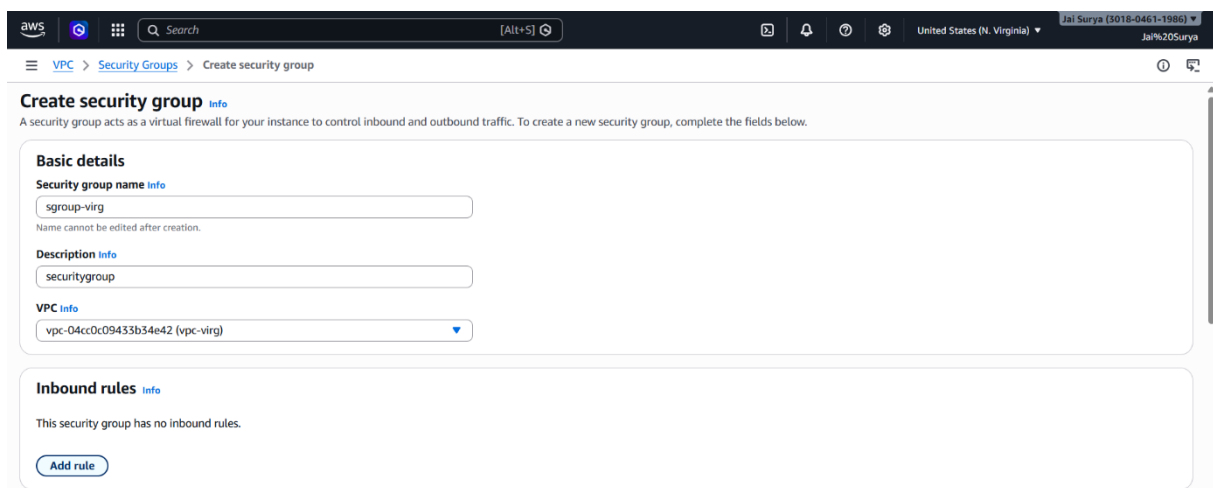


Screenshot – Security Group Details

Step 1: Enter Name and Description.

Step 2: Select the VPC.

Step 3: Click Add rule in Inbound rules.

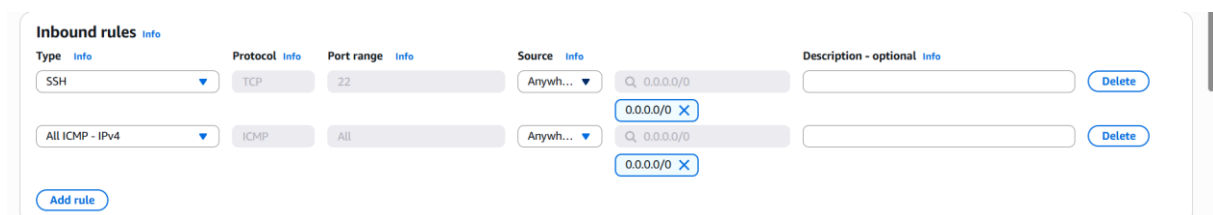


Screenshot – Add Inbound Rules

Step 1: Add SSH rule.

Step 2: Add All ICMP – IPv4 rule.

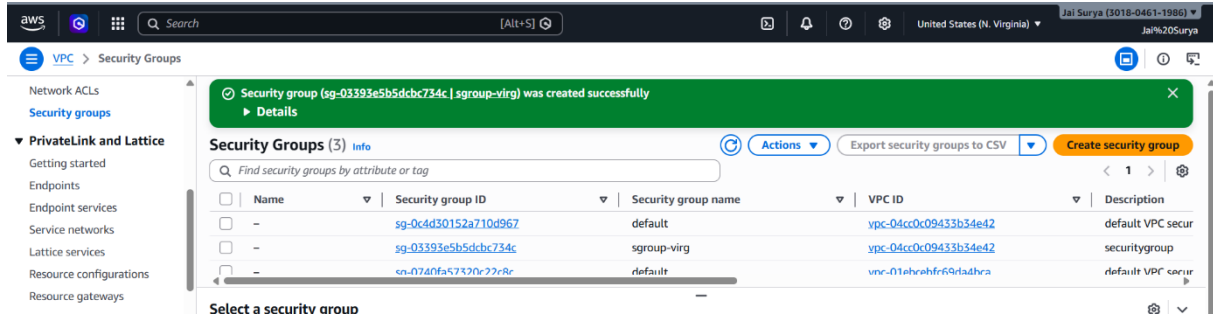
Step 3: Set Source as Anywhere.



Screenshot – Security Group Created

Step 1: Click Create security group.

Step 2: Security group is created successfully.



Screenshot – Creating Linux Instance in Virginia

Step 1: Enter Name.

Step 2: Select Linux AMI.

Step 3: Select Key pair.

Step 4: Click Edit in Network settings.

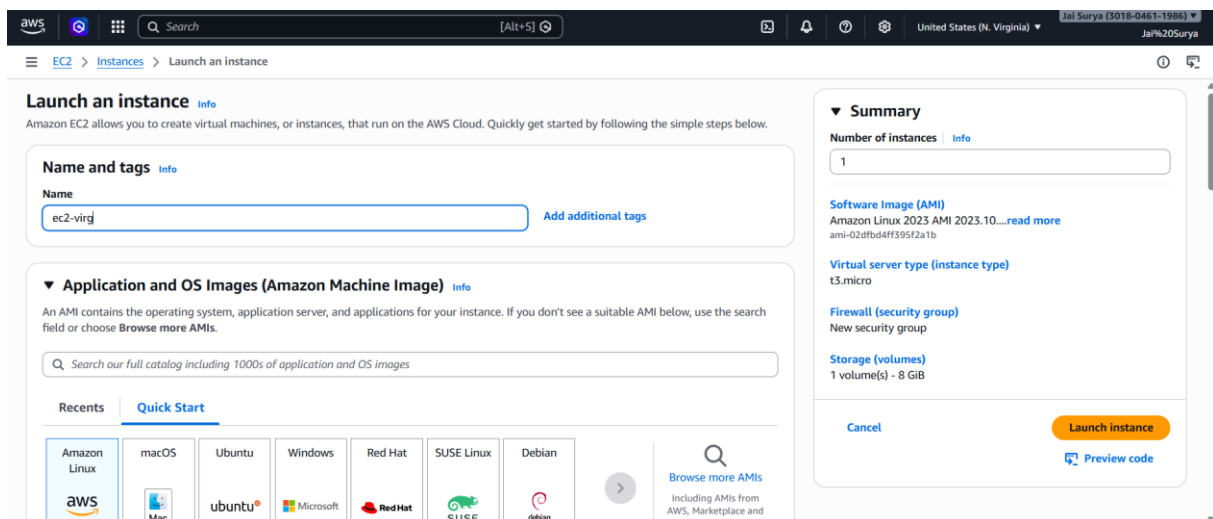
Step 5: Select VPC and Subnet.

Step 6: Enable Auto-assign Public IP.

Step 7: Select Existing security group.

Step 8: Choose the created security group.

Step 9: Click Launch instance.



▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

virg

Create new key pair

EC2 > Instances > Launch an instance

▼ Network settings Info

VPC - required Info

vpc-04cc0c09433b34e42 (vpc-virg)

10.0.0.0/16

Subnet Info

subnet-0eee099df5a491aae subnet-virg

VPC: vpc-04cc0c09433b34e42 Owner: 301804611986 Availability Zone: us-east-1a (use1-az1)

Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Common security groups Info

Select security groups

sgroup-virg sg-03395e5b5dbc734c

VPC: vpc-04cc0c09433b34e42

Compare security group rules

▼ Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...read more

ami-02dfbd4ff395f2a1b

Virtual server type (instance type)

t3.micro

Firewall (security group)

sgroup-virg

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Screenshot – Instance Created

Step 1: Click Launch instance.

Step 2: Instance is created successfully.

aws

Search

[Alt+S]

United States (N. Virginia)

Jai Surya (3018-0461-1986)

Jai%20Surya

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

Instances

Images

Instances (1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	ec2-virg	i-0d11f500091d2b4d8	Running	t3.micro	Initializing	View alarms +	us-east-1a	-

CREATE RESOURCES IN SINGAPORE REGION

Screenshot – Create Resources in Singapore Region

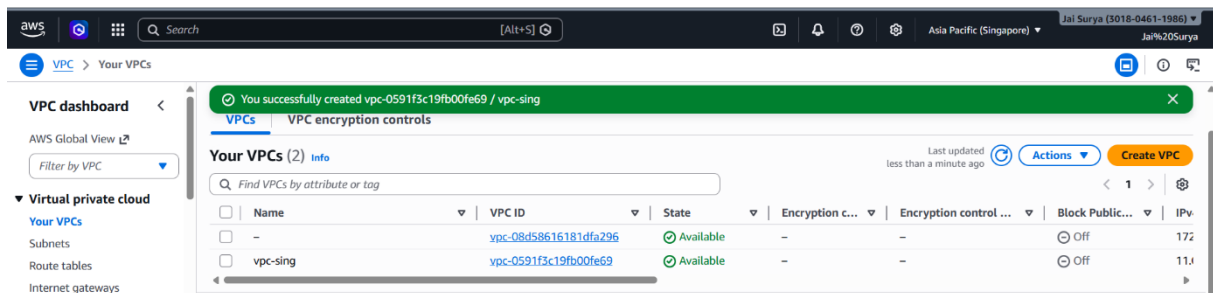
Step 1: Change region to Singapore.

Step 2: Create VPC, Subnet, Route Table, Internet Gateway, and Security Group.

Step 3: Launch a Linux instance with the same settings.

Step 4: Resources are created successfully in Singapore region.

VPC



SUBNET

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

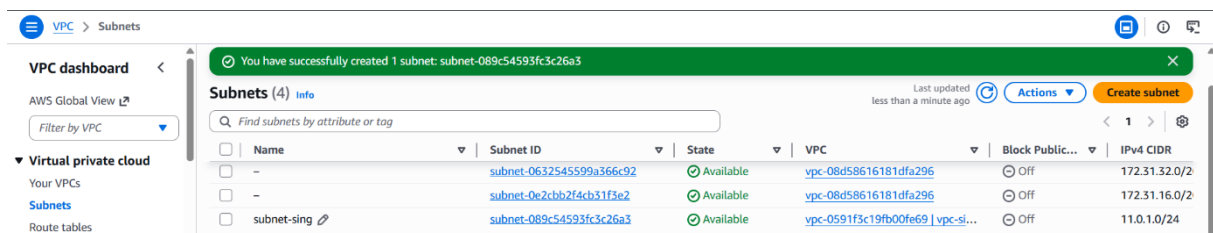
Subnet name
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block
 256 IPs



Route table

Updated routes for rtb-0c5e2e8b822444ca5 / rt-sing successfully

Route tables (1/3) Info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-00e2529edc033f497	-	-	Yes	vpc-08d58616181dfa296
rt-sing	rtb-0c5e2e8b822444ca5	subnet-089c54593fc3c26...	-	No	vpc-0591f3c19fb00fe69 vpc-

rtb-0c5e2e8b822444ca5 / rt-sing

Internet gateway

Internet gateway igw-0b6cf9efabd05554e successfully attached to vpc-0591f3c19fb00fe69

Internet gateways (1/2) Info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-00fa2e8b471f201d3	Attached	vpc-08d58616181dfa296	301804611986
internet-sing	igw-0b6cf9efabd05554e	Attached	vpc-0591f3c19fb00fe69 vpc-sing	301804611986

Security group

Security group (sg-01621de02c8268d36 | sgroup-sing) was created successfully

Security Groups (3) Info

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description
-	sg-064f163affdd8e56a	default	vpc-0591f3c19fb00fe69	default VPC secur
-	sg-0cb9a44c1113d37fe	default	vpc-08d58616181dfa296	default VPC secur
-	sg-01621de02c8268d36	sgroup-sing	vpc-0591f3c19fb00fe69	sgroup-sing

Select a security group

Instance in singapore

Instance i-0995a181c7d118e53 was created successfully

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

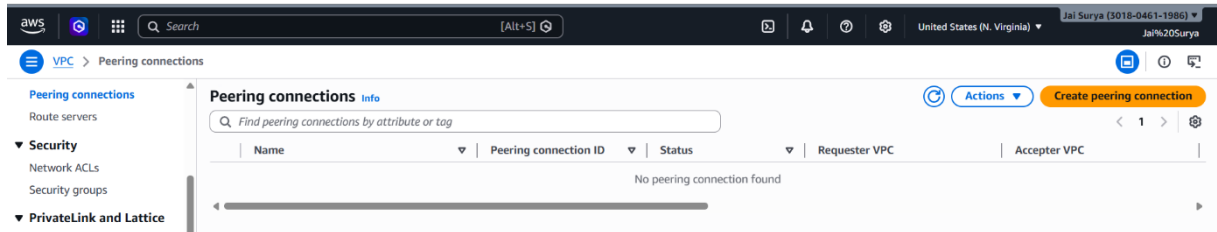
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
ec2-sing	i-0995a181c7d118e53	Running	t3.micro	Initializing	View alarms +	ap-southeast-1a	-

PEERING CONNECTION

Screenshot – Create Peering Connection

Step 1: Go to Peering Connections.

Step 2: Click Create peering connection.



Screenshot – Peering Connection Details

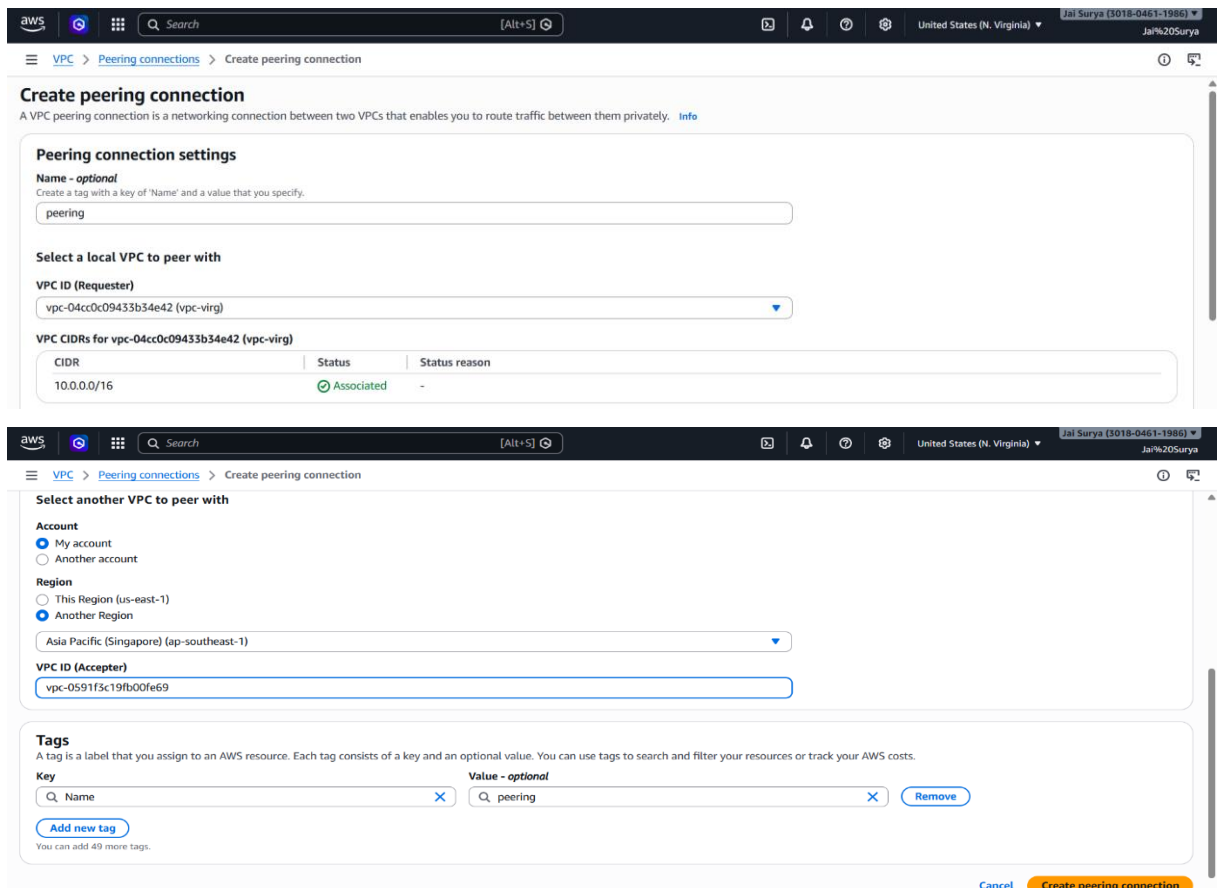
Step 1: Enter Peering connection name.

Step 2: Select Virginia VPC as Requester.

Step 3: Select My account.

Step 4: Choose Another region.

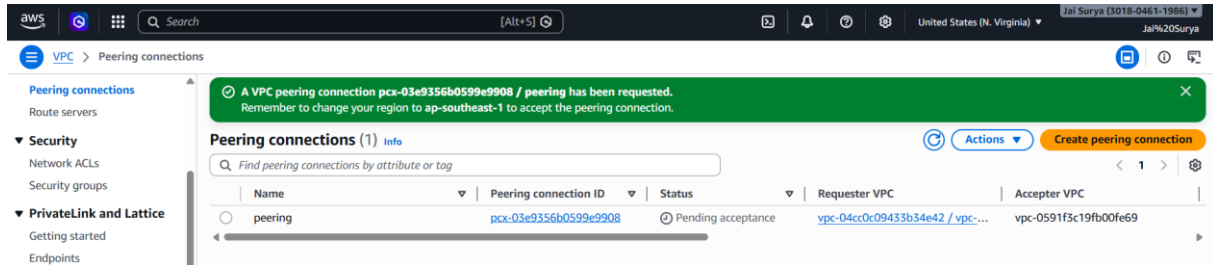
Step 5: Enter Singapore VPC ID in Acceptor.



Screenshot – Peering Request Sent

Step 1: Peering connection request is sent.

Step 2: Status shows Pending Acceptance in Virginia.



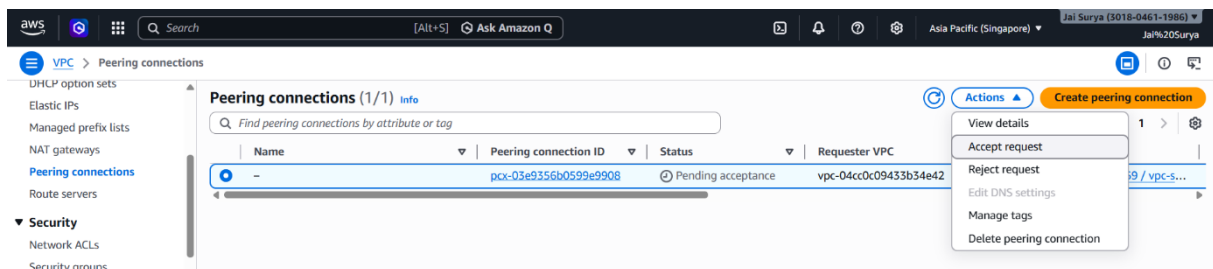
Screenshot – Accept Peering Request

Step 1: Go to Singapore region.

Step 2: Go to Peering Connections.

Step 3: Select the peering connection.

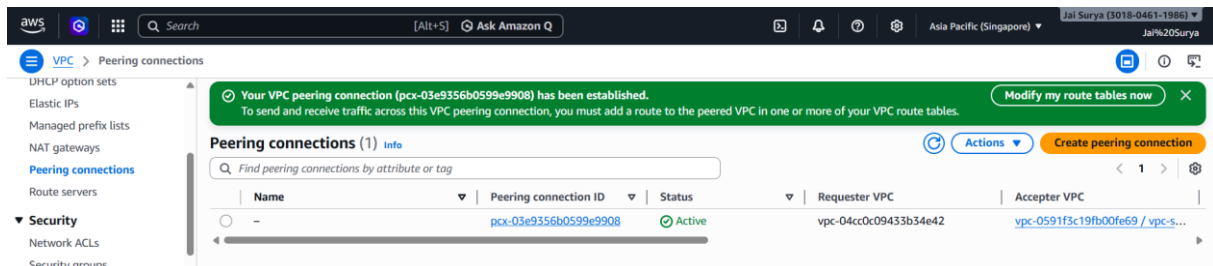
Step 4: Click Actions → Accept request.

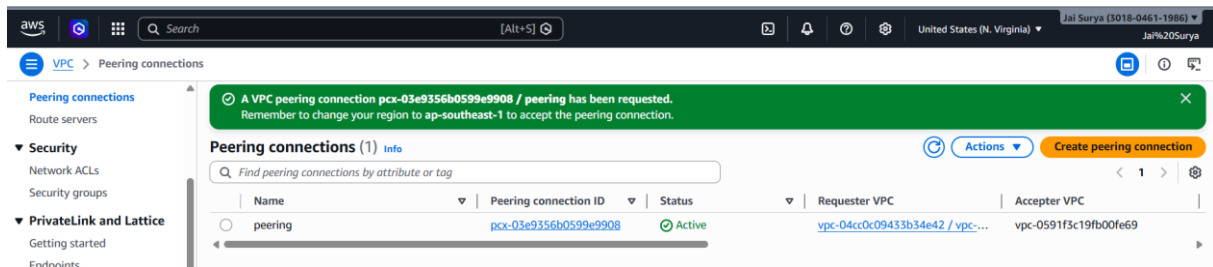


Screenshot – Peering Connection Active

Step 1: Check Peering Connections.

Step 2: Status shows Active in both regions.





Screenshot – Add Route for Peering in Virginia

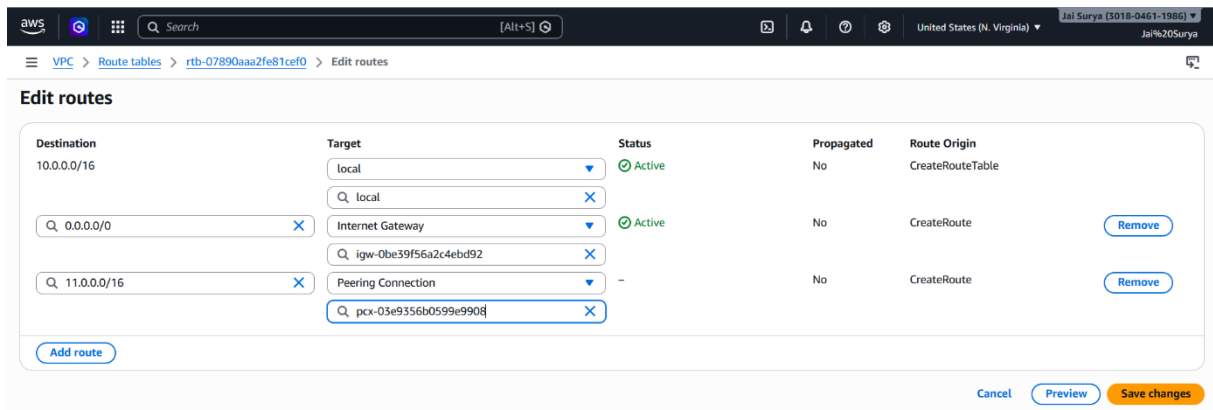
Step 1: Go to Route Tables in Virginia.

Step 2: Open Routes and click Edit.

Step 3: Add Singapore VPC CIDR.

Step 4: Select Peering connection as target.

Step 5: Choose the Peering ID and click Save.



Screenshot – Add Route for Peering in Singapore

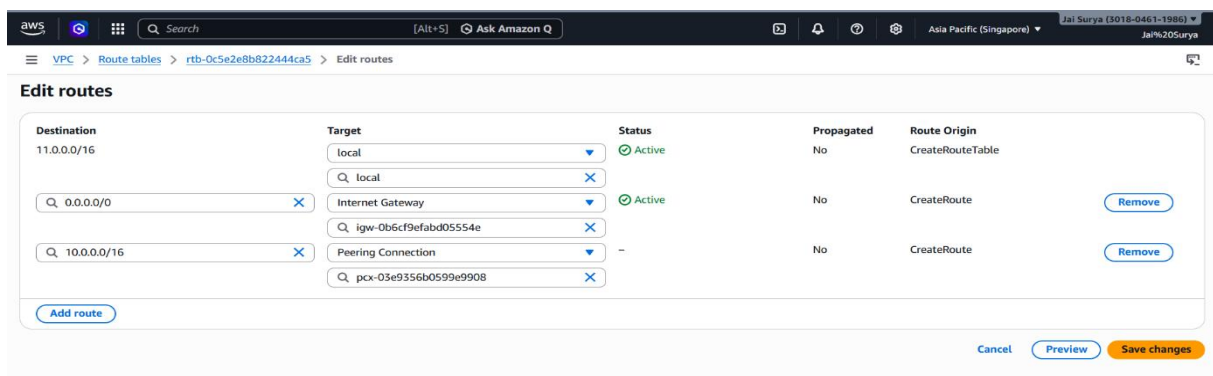
Step 1: Go to Route Tables in Singapore.

Step 2: Open Routes and click Edit.

Step 3: Add Virginia VPC CIDR.

Step 4: Select Peering connection as target.

Step 5: Choose the Peering ID and click Save.

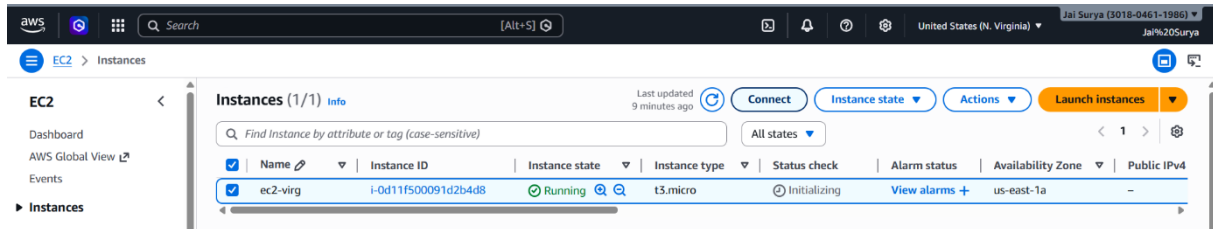


Screenshot – Connect Both Instances

Step 1: Open Instances.

Step 2: View Virginia instance and Singapore instance.

Step 3: Both instances are running.



Screenshot – Ping Test

Step 1: Open Virginia Linux terminal.

Step 2: Run ping <private IP of Singapore instance>.

Step 3: Ping is successful, showing connectivity.



Screenshot – Ping Test

Step 1: Open Singapore Linux terminal.

Step 2: Run ping <private IP of Virginia instance>.

Step 3: Ping is successful, showing connectivity.

